TateLedger: Public Mathematical Ledger of the Tate Conjecture, Known Cases, Implications, and Hodge Comparison

Benjamin Stanley Frohman
ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/TateLedger
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Abstract

Standalone reference ledger (‘docs/STATEMENT.md‘, ‘docs/KNOWN_CASES.md‘, ‘docs/IMPLICATIONS.m
‘docs/HODGE_COMPARISON.md‘, ‘docs/NOT_THIS_HOST.md‘) cataloging the exact scope of the Tate Conjecture, proven cases (Faltings for divisors on abelian varieties over number fields,Zarhin over function fields, Charles-Madapusi Pera-Maulik for K3 surfaces), implications for BSD/Mordell-Weil ranks, and structural differences from the complex Hodge Conjecture.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Reference ledger cataloging literature results and open problems; explicitly does not claim a proof of the general Tate or Hodge conjectures.

- **Repository:** `https://github.com/BenFrohman/TateLedger`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23652982`
- **Earliest Git Commit SHA:** `509539734107143 64f5815a21897dcb594b93bf6` (2026-09-28 05:16:55 AM CDT (UTC-5))
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2 Technical Overview & Repository Specification

Author: Benjamin Stanley Frohman ([@BenFrohman](https://github.com/BenFrohman))
Copyright: Copyright (c) 2026 Benjamin Stanley Frohman
License: [Apache-2.0](LICENSE)
Public ledger of the Tate conjecture.
This repository is **not** a proof of Tate.
This repository is **not** a proof of Hodge.
It names the sentence, records implications, and points at prior theorems.
Lean skeleton of the same sentence: [BenFrohman/TateConjecture](https://github.com/BenFrohman/TateC
Complex Hodge ledger: [BenFrohman/HODGE](https://github.com/BenFrohman/HODGE)
Let ‘X‘ be smooth projective over a finite field ‘F_q‘.
After a Tate twist, the cycle class map lands in Galois invariants:

cl : CH^i(X)_Q \{}otimes Q_\{}ell \{}to H^{2i}_et(X_{bar F_q}, Q_\{}ell(i))^{Gal}
Tate: that map is surjective.

The numerical shadow is that geometric Frobenius on ‘H^{2i}‘ has eigenvalue ‘q^i‘ on the Tate classes. For fourfolds and ‘i = 2‘ that shadow is ‘q^2‘ on ‘H^4‘. Weil already forces ‘|\{}alpha| = q^2‘ on ‘H^4‘. The root ‘q^2‘ is a condition on a class, not a surface.

Writing ‘cl(z) = \{}gamma‘ is the hard arrow. This repository does not write those cycles for an unspecified ‘X‘.

File	Content
—	—
docs/STATEMENT.md	the sentence

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/TateLedger` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTIFICA` and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
`CONTRIBUTING.md`	1061 B	`435ecbc4472f43dd...`
`COPYRIGHT.md`	647 B	`593f626c18539dfd...`
`LICENSE`	11355 B	`9da72c2bee4f20ba...`
`NOTICE`	322 B	`35405ec5b9c57966...`
`NOTICE.md`	260 B	`f80731613f4432cb...`
`README.md`	2208 B	`5d3dc7e194a117d9...`
`SECURITY.md`	931 B	`3ef25a0925599cd7...`
`docs/HODGE_COMPARISON.md`	1028 B	`f599e80f2b6903bf...`
`docs/IMPLICATIONS.md`	1655 B	`6588e9641ed453fa...`
`docs/KNOWN_CASES.md`	1192 B	`0cf97fc745556183...`
`docs/LITERATURE.md`	1663 B	`175d23a8d6081769...`
`docs/NOT_THIS_HOST.md`	910 B	`08847092f23f1684...`
`docs/STATEMENT.md`	1163 B	`b58af703a20ea4e3...`

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